Model Predictive Control | Spring 2026
Prof. M. Zeilinger

Exercise 04
CFTOC

Exercise 1 **MPC Problem with Quadratic Costs**

Consider the optimization problem with parameter $x(0)$:

$$
\begin{aligned}
f^*(x(0)) = \min \frac{1}{2}\left(\sum_{i=0}^{2} x_i^2 + \sum_{i=0}^{1} u_i^2\right) && \text{(4a)}\\
\text{subject to} \quad & x_0 = x(0) && \text{(4b)}\\
& x_{i+1} = 0.5x_i + u_i,\ i \in \{0,1\} && \text{(4c)}\\
& 2.5 \leq x_1 \leq 5 && \text{(4d)}\\
& -1 \leq x_2 \leq 1 && \text{(4e)}\\
& -2 \leq u_0 \leq 2 && \text{(4f)}\\
& -2 \leq u_1 \leq 2 && \text{(4g)}
\end{aligned}
$$

for a SISO process with linear dynamics (4c) given in state space representation, state constraints (4d)-(4e) and input constraints (4f)-(4g). The initial state x_0 is defined by the equality constraint (4b) and is therefore identical to the parameter $x(0)$.

1. Let $x(0) = 2$. Solve the QP using the MATLAB function `quadprog`. For this it is convenient to first convert the QP into the format required by `quadprog` with decision vector x defined appropriately:

$$
\begin{aligned}
\min \ & \frac{1}{2}x^T Hx + f^T x\\
\text{s.t.} \ & Ax \leq b\\
& A_{eq}x = b_{eq}
\end{aligned}
$$

   Note: This format is also useful for checking the KKT conditions in the next task in a more convenient way.

2. Check the KKT conditions: Do they hold? Which constraints are active?
   Hint: `quadprog` returns the vector of Lagrangian multipliers if invoked with a suitable number of output arguments (cf. `help quadprog`).

3. Plot the minimizer at time step 0 (i.e. the optimal control input u_0^*) and the minimum f^* as a function of the parameter $x(0)$. What do you observe?

4. For $x(0) = 2$, how would minimum and minimizer change if we removed the lower bound on x_1 (i.e., removed constraint $x_1 \geq 2.5$)?
   What if we removed the upper bound on x_1 (i.e., removed constraint $x_1 \leq 5$)? Explain your answers.

Now consider the same problem with a 1-norm cost function

$$
\begin{aligned}
f^*(x(0)) = & \min \left(\sum_{i=0}^{2} |x_i| + \sum_{i=0}^{1} |u_i| \right) && \text{(5a)} \\
\text{subject to} \quad & x_0 = x(0) && \text{(5b)} \\
& x_{i+1} = 0.5 x_i + u_i, \; i \in \{0, 1\} && \text{(5c)} \\
& 2.5 \le x_1 \le 5 && \text{(5d)} \\
& -1 \le x_2 \le 1 && \text{(5e)} \\
& -2 \le u_0 \le 2 && \text{(5f)} \\
& -2 \le u_1 \le 2, && \text{(5g)}
\end{aligned}
$$

where the cost function is linear.

5. Let $x(0) = 2$. Solve the LP using the MATLAB function `linprog`. For this it is convenient to first convert the LP into the format required by `linprog` with decision vector x defined appropriately:

$$
\begin{aligned}
& \min f^T x \\
\text{s.t. } & Ax \le b \\
& A_{eq} x = b_{eq}
\end{aligned}
$$

6. Plot the minimizer at time step 0 (i.e. the optimal control input u_0^*) and the minimum f^* as a function of the parameter $x(0)$. What do you observe?

7. For $x(0) = 2$, how would minimum and minimizer change if we removed the lower bound on x_1 (i.e., removed constraint $x_1 \ge 2.5$)?
   What if we removed the upper bound on x_1 (i.e., removed constraint $x_1 \le 5$)? Explain your answers.

Exercise 2 **1-Norm, ∞-Norm**

Instead of the standard 2-norm, the 1-norm or the ∞-norm are sometimes used in MPC cost functions. In this exercise, you should show that both a minimization problem with a 1-norm objective and an ∞-norm objective

$$
\min_x \; \|Ax\|_p \,, \qquad p \in \{1, \infty\}
$$

can be recast as a linear program (LP)

$$
\begin{aligned}
& \min_y \; b^T y \\
\text{s.t. } & Fy \le g \;\; ,
\end{aligned}
$$

with vectors b, g and matrix F defined appropriately. Note that we assume matrix $A \in \mathbb{R}^{N \times n}$ where its N row vectors are denoted as a_i^T, $i = 1, \dots N$, i.e.

$$
A = \begin{bmatrix} a_1^T \\ \vdots \\ a_N^T \end{bmatrix} .
$$